

1. Suppose that $f : [0, 1] \rightarrow [0, 1]$ is continuous. Prove that there exists $x \in [0, 1]$ such that $f(x) = x$.
2. Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous functions, and suppose that $f(a) > g(a)$ and $f(b) < g(b)$. Prove that $f(c) = g(c)$ for some $c \in [a, b]$.
3. Use the definition of derivative to find the derivative of $f(x) = x^3$ at $a \in \mathbb{R}$.

Definition 1. A set $A \subset \mathbb{R}$ is *connected* if whenever $a, b \in A$ and $a < c < b$, then $c \in A$. Otherwise, A is said to be *disconnected*.

4. Show that a set $A \subset \mathbb{R}$ is disconnected if and only if there exists open sets P and Q such that
 - $P \cap Q = \emptyset$ and $A \subset P \cup Q$
 - $P \cap A \neq \emptyset$ and $Q \cap A \neq \emptyset$
- (a) $(\Rightarrow)$
- (b) $(\Leftarrow)$